Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	PE of system / planet = [-]$1 \times 10^{10} \times 6.2 \times 10^{25}$ = (6.2×10^{35} J) (1) **Alternative for the 1st mark:** Obtaining escape velocity: 140 000 [m s^{-1}] **and** velocity of 7×10^{35} [J] i.e. 150 000 [m s^{-1}] **Another alternative for the 1st mark:** $\frac{7 \times 10^{35}}{6.2 \times 10^{25}} = 1.13 \times 10^{10}$ [J kg^{-1}] However, conservation of momentum / other energy transfers must be considered (1) minimum required is {elastic / inelastic} collision Conclusion consistent with their argument but must be based on a value of PE **or** potential **or** escape velocity (1)			3	3	1	
	(d)		Star is at centre of mass **or** resultant force will be zero (1) Hence, no wobble / orbit etc. **or** stationary (1) No Doppler shift and not detectable so Delyth is correct (1)			3	3		
			Question 3 total	**2**	**9**	**8**	**19**	**10**	**0**